

Chen Yang

World Models Researcher & Engineer

Principal Engineer at HUAWEI

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Education

- Sep 2021 – Jun 2025 📖 **Ph.D. in Computer Science, Shanghai Jiao Tong University**
Ph.D. Advisor: **Prof. Wei Shen**
Laboratory: **Key Lab of Artificial Intelligence, Ministry of Education**
- Sep 2019 – Jun 2021 📖 **M.A. in Precision Instrument, Shanghai Jiao Tong University**
GPA: 3.76/4.0 | Top 10%
- Sep 2015 – Jun 2019 📖 **B.A. in Precision Instrument, Shanghai Jiao Tong University**
GPA: 3.91/4.3 | Top 30%

Selected Research Projects

- 2024 📖 **LiftImage3D: Lifting Any Single Image to 3D Gaussians with Video Generation Priors**
★ LiftImage3D is a framework that effectively releases the generative priors of latent video diffusion models (LVDMs) while ensuring 3D consistency; can lift arbitrary 2D images into high-quality 3D scenes composed of 3D Gaussians.
★ Project lead by **Dr. Jiemin Fang and Dr. Qi Tian**.
- 2023 – 2024 📖 **GaussianObject: High-Quality 3D Object Reconstruction from Four Views with Gaussian Splatting**
★ Aiming to reconstruct finely detailed objects from very sparse inputs (as few as 4 images). Leveraging 3DGS as scene representation and refining a pre-trained diffusion model for strong priors.
★ Project lead by **Prof. Wei Shen and Dr. Jiemin Fang**.
★ Accepted by **ACM Transactions on Graphics (TOG), SIGGRAPH Asia 2024**.
- 📖 **EndoGSLAM: Real-Time Dense Reconstruction and Tracking in Endoscopic Surgeries using Gaussian Splatting**
★ Developing a novel SLAM approach for endoscopic surgeries that achieves over 100 fps rendering speed during online camera tracking and tissue reconstruction using streamlined 3D Gaussians.
★ Project lead by **Prof. Wei Shen**.
★ Accepted by **MICCAI 2024**.
- 2023 📖 **Segment Anything in 3D with NeRFs**
★ Leveraging SAM (Segment Anything) to segment NeRFs, provide a generic method to lift 2D foundation models to the 3D space.
★ Project lead by **Dr. Jiemin Fang and Prof. Wei Shen**.
★ Accepted by **NeurIPS 2023**.

Selected Research Projects (continued)

- 2022 – 2023  **Efficient Deformable Tissue Reconstruction via Orthogonal Neural Plane**
★ Accelerated the optimization and inference on reconstructing deformable tissues with NeRFs, improving efficiency and quality across non-rigid deformations.
★ Project lead by **Prof. Wei Shen**.
★ Accepted by **MICCAI 2023, Young Scientist Award** and **IEEE Transactions on Medical Imaging (TMI)**. Inspired **20+ publications** on dynamic tissue reconstruction.
- 2021 – 2022  **NeRFVS: Neural Radiance Fields for Free View Synthesis via Geometry Scaffolds**
★ Designed a novel approach enabling neural radiance fields to perform free view synthesis at room scale and perform superior extrapolation in room scale.
★ Project lead by **Dr. Weichao Qiu** and **Prof. Wei Shen**.
★ Accepted by **CVPR 2023**.

Internship Experience

- 2023 – 2024  **3D Vision Intern, Huawei Cloud**, mentored by **Dr. Jiemin Fang** and **Dr. Qi Tian**
★ Designed and implemented GaussianObject project, which enables high-quality 3D object reconstruction from very sparse inputs (as few as 4 images);
★ Paper accepted by SIGGRAPH Asia 2024 (TOG), work is open-sourced and widely recognized (**1,200+ GitHub Stars**).
- 2021 – 2022  **Machine Vision Intern, Huawei Noah's Ark Lab**, mentored by **Dr. Weichao Qiu**
★ Designed and implemented NeRFVS project, significantly improving the extrapolation capability of neural radiance fields;
★ Proposed geometry scaffolds method, substantially enhancing scene reconstruction quality and extrapolation performance, paper accepted by CVPR 2023.

Awards and Achievements

- 2023  **MICCAI Young Scientist Award**, Awarded top 5 among 2250 submissions.
  **Intel Scholarship**, Awarded top 5 among over 100 competitors.
- 2022  **Second Prize of National Post-Graduate Mathematical Contest in Modeling**, Awarded to top 14.5% of contestants.
- 2021  **National Scholarship**, Awarded to top 3% of students at Shanghai Jiao Tong University.
  **First Prize of Huawei Chinese University ICT Competition**, Awarded top 1 among 88 teams.
- 2019 – 2021  **First-class Academic Scholarship**, Awarded to top 30% of students at Shanghai Jiao Tong University.

Research Publications

- 1 **C. Yang***, S. Li*, J. Fang, et al., "Gaussianobject: Just taking four images to get a high-quality 3d object with gaussian splatting," **ACM TOG**, 2024.
- 2 **C. Yang**, K. Wang, Y. Wang, X. Yang, and W. Shen, "Neural lerplane representations for fast 4d reconstruction of deformable tissues," in **MICCAI, Young Scientist Award**, Springer Nature Switzerland Cham, 2023, pp. 46–56.

- 3 **C. Yang**, P. Li, Z. Zhou, et al., "Nerfvs: Neural radiance fields for free view synthesis via geometry scaffolds," in *CVPR*, 2023, pp. 16 549–16 558.
- 4 **C. Yang**, K. Wang, Y. Wang, Q. Dou, X. Yang, and W. Shen, "Efficient deformable tissue reconstruction via orthogonal neural plane," *IEEE TMI*, 2024.
- 5 **C. Yang**^{*}, K. Wang^{*}, Y. Wang, et al., "Endogslam: Real-time dense reconstruction and tracking in endoscopic surgeries using gaussian splatting," in *MICCAI*, Springer, 2024, pp. 219–229.
- 6 **C. Yang**, S.-Y. Yao, Z.-W. Zhou, B. Ji, G.-T. Zhai, and W. Shen, "Poxture: Human posture imitation using neural texture," *IEEE TCSVT*, vol. 32, no. 12, pp. 8537–8549, 2022.
- 7 **C. Yang**^{*}, H. Zhao^{*}, H. Wang, and W. Shen, "Chase: 3d-consistent human avatars with sparse inputs via gaussian splatting and contrastive learning," arXiv preprint arXiv:2408.09663, 2024.
- 8 K. Wang^{*}, **C. Yang**^{*}, K. Zhao, X. Yang, and W. Shen, "Realistic surgical simulation from monocular videos," arXiv preprint, 2024.
- 9 Y. Chen^{*}, **C. Yang**^{*}, J. Fang, et al., "Liftimage3d: Lifting any single image to 3d gaussians with video generation priors," arXiv preprint, 2024.
- 10 H. Zhao^{*}, **C. Yang**^{*}, H. Wang, X. Zhao, and W. Shen, "Sg-gs: Photo-realistic animatable human avatars with semantically-guided gaussian splatting," arXiv preprint arXiv:2408.09665, 2024.
- 11 B. Ji, **C. Yang**, Y. Shunyu, and Y. Pan, "Hpof: 3d human pose recovery from monocular video with optical flow," in *ICMR*, 2021, pp. 144–154.
- 12 P. Li, S. Wang, **C. Yang**, B. Liu, W. Qiu, and H. Wang, "Nerf-ms: Neural radiance fields with multi-sequence," in *ICCV*, 2023, pp. 18 591–18 600.
- 13 J. Cen, J. Fang, **C. Yang**, et al., "Segment any 3d gaussians," in *AAAI*, 2024.
- 14 R. Liang, J. Zhang, H. Li, **C. Yang**, Y. Guan, and N. Vijaykumar, "Spidr: Sdf-based neural point fields for illumination and deformation," *CVPRW*, 2023.
- 15 Y. Yan, Z. Zhou, Z. Wang, **C. Yang**, J. Gao, and X. Yang, "Dialoguenerf: Towards realistic avatar face-to-face conversation video generation," *Visual Intelligence*, vol. 2, no. 1, p. 24, 2024.
- 16 J. Cen, Z. Zhou, J. Fang, **C. Yang**, et al., "Segment anything in 3d with nerfs," *NeurIPS*, vol. 36, pp. 25 971–25 990, 2023.
- 17 J. Lu, T. Yi, J. Fang, **C. Yang**, et al., "Snap-snap: Taking two images to reconstruct 3d human gaussians in milliseconds," arXiv preprint, 2024.
- 18 Z. Zhou, R. Zhong, **C. Yang**, Y. Wang, X. Yang, and W. Shen, "A k-variate time series is worth k words: Evolution of the vanilla transformer architecture for long-term multivariate time series forecasting," arXiv preprint arXiv:2212.02789, 2022.

Skills

Programming Languages	Python, C, C++, Matlab
Software & Tools	PyTorch, OpenCV, OpenGL, L ^A T _E X, Jax, COMSOL

Academic Services

Conference Reviewer	CVPR '23, '24; ICCV '23; NeurIPS '23, '24; ECCV '24; AAAI '25; MICCAI '23, '24
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Academic Services (continued)

Journal Reviewer	■ TOG; TMI; TCSVT; TOMM
Teaching Assistant	■ Spr. 2019: MI 321 : Course Design of Instrument Bus and Virtual Env. ★ Guided students in bus programming and virtual instrument development projects. Fa. 2020: MI 318 : Measuring and Controlling Circuit ★ Facilitated lab sessions and assisted students with hands-on circuit experiences. Spr. 2021: EE 334 : Industrial Measurement and Control Tech. and Sys. ★ Guided students through industrial measurement processes and control technologies.